

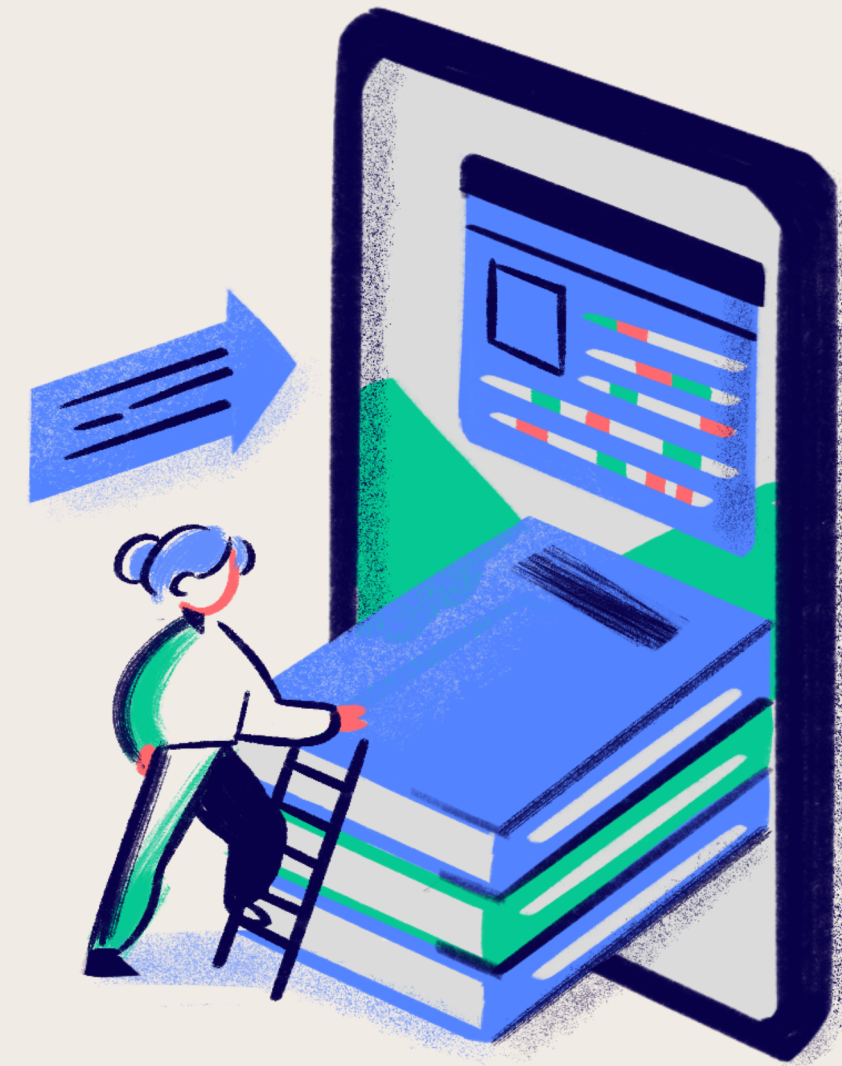
TITLE: SIMULATION OF COAP PROTOCOL IN IOT NETWORKS USING COOJA: A LIGHTWEIGHT AND SCALABLE APPROACH

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ABSTRACT

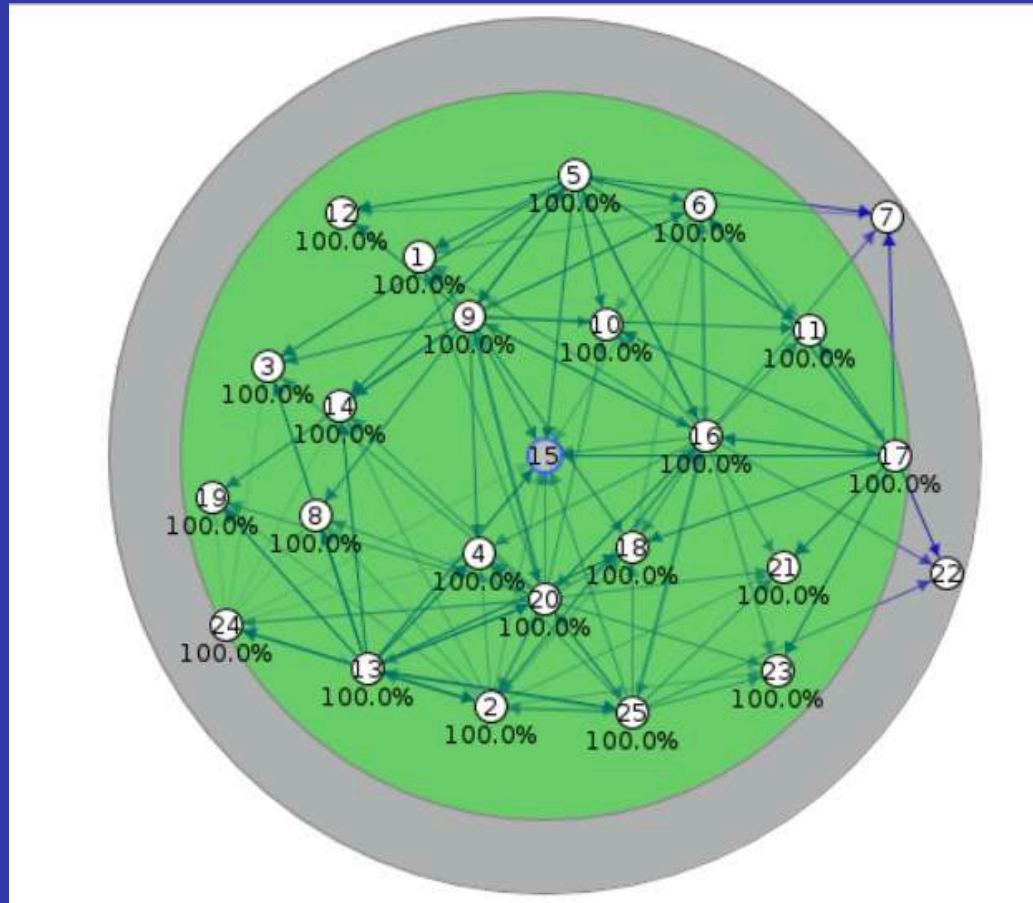
- This project focuses on simulating the Constrained Application Protocol (CoAP) using the Cooja simulator integrated with the Contiki-NG OS.
- CoAP is designed for lightweight, constrained IoT devices and is widely adopted for machine-to-machine (M2M) communication.
- The aim is to simulate a resource-constrained network using CoAP and evaluate its latency, packet delivery ratio, and energy consumption.
- The Cooja simulator enables emulation of real sensor nodes and supports detailed performance evaluation in wireless sensor networks (WSNs).
- This simulation will help us understand how CoAP behaves under various conditions, aiding in protocol selection and optimization in IoT systems.
- The project is executed using a single open-source tool (Cooja), making it suitable for implementation on a personal laptop without additional hardware or commercial software.
- Our deliverables include: simulation setup, traffic analysis, performance graphs, and interpretation of results.

ABOUT THE PROTOCOL

- CoAP (Constrained Application Protocol) is a lightweight application layer protocol designed specifically for IoT and resource-constrained devices.
- It follows a RESTful model similar to HTTP but operates over UDP to reduce overhead.
- Supports methods like GET, POST, PUT, DELETE, enabling standard CRUD operations in IoT communication.
- Built-in support for asynchronous communication, multicast, and retransmissions for reliability.
- Often used in smart homes, environmental monitoring, industrial automation, etc.
- Works efficiently in low-power and lossy networks (LLNs) commonly found in wireless sensor environments.
- Key feature: low header overhead and binary message format, making it ideal for small devices.



ABOUT THE TOOL



- Cooja is a flexible network simulator part of the Contiki-NG operating system, widely used for Wireless Sensor Network (WSN) and IoT protocol testing.
- Simulates real motes (e.g., Sky, Z1) and allows programming them with Contiki OS firmware.
- Allows users to visualize topologies, transmission ranges, node behaviors, and energy consumption.
- Supports protocol-level packet inspection, node-to-node communication, and traffic monitoring.
- Built-in tools for simulation control, packet logs, and radio message tracking.
- Can be run on any basic system with Java, making it ideal for personal laptop setups without external hardware.
- Frequently used in academic research to test, visualize, and analyze new protocols in realistic IoT environments.

LITERATURE SURVEY

#	Title	Authors	Year	Tool / Platform	Focus / Protocol Studied	Key Findings	Research Gap / Justification
1	Performance Evaluation of Advanced Congestion Control Mechanisms for CoAP	Sanaboina & Eluri	2023	Contiki OS, COOJA	CoAP (Default, CoCoA, CoCoA+)	CoCoA+ performs best in delivery ratio and latency under constant traffic	Mobility, heterogeneous traffic patterns not considered
2	Performance analysis of constrained application protocol using Cooja simulator in Contiki OS	Naik & Joshi	2017	COOJA, Contiki OS	CoAP	Analyzed delay, throughput, and delivery across scenarios	Memory overhead and large-scale scenarios unexplored
3	Power Evaluation of IoT Application Layer Protocols	Shahrokhi & Ahmadi	2024	Contiki, COOJA + Powertrace	CoAP vs MQTT-SN vs HTTP	MQTT-SN consumes least power; CoAP better than HTTP	Limited to power evaluation, no congestion analysis

4	Performance Evaluation of CoAP and MQTT-SN in an IoT Environment	Martí et al.	2019	Contiki, COOJA	CoAP vs MQTT-SN	CoAP lower latency; MQTT-SN more lightweight	Doesn't assess reliability or large topologies
5	Performance Evaluation of RPL, CoAP and 6LoWPAN Protocol for Different Time Interval and PDR in IoT Environment	Kaptan et al.	2019	COOJA + Contiki	CoAP with RPL + 6LoWPAN	Increased node count reduces PDR	No energy, mobility, or delay analysis
6	Performance Analysis of an IP based Protocol Stack for WSNs	Thombre, et al.	2015	COOJA + Contiki	CoAP + 6LoWPAN	Deeper networks show more packet loss and delay	Security and adaptive routing not evaluated
7	On the Design of Efficient Congestion Control for the Constrained Application Protocol (CoAP)	Aujla et al.	2022	COOJA, Contiki	IDC-CoAP, MBC-CoAP	Improved timeout and congestion control vs CoCoA+	Further evaluation under burst traffic needed

8	Congestion Control in Reliable CoAP Communication	Bormann et al.	2013	Contiki + COOJA	Reliable CoAP	Efficient lightweight congestion control proposed	Real-world interoperability not validated
9	Design and Implementation of an Antenna Model for the Cooja Simulator	Rege	2015	COOJA	Antenna modeling for CoAP	Enables directional radio evaluation with CoAP	Integration with dynamic topologies needed
10	Performance Evaluation and Analysis of LightCert4IoT using Cooja-Contiki Simulator	Khoury et al.	2024	COOJA + Contiki	CoAP with lightweight certificate bootstrapping	Acceptable energy and memory cost for secure provisioning	No attack-resistance testing or dynamic load validation

JUSTIFICATION FOR CHOOSING THIS TOPIC

- **Lightweight Protocol Focus:**
- Many papers show that CoAP is ideal for constrained devices due to its low overhead compared to HTTP and MQTT, making it crucial to simulate and optimize further.
- **Simulation-Oriented Research Gaps:**
- Despite several studies using COOJA, most focus only on specific metrics (e.g., energy or delivery) and don't holistically assess scalability, traffic variation, and reliability together.
- **Scalability Needs Exploration:**
- As seen in Paper #6 and #5, performance degrades in deep or dense topologies, but few studies propose generalizable or scalable solutions—hence the need for a dedicated approach.
- **Power & Efficiency Trade-offs:**
- Studies like Paper #3 reveal CoAP is energy-efficient but still underexplored compared to MQTT-SN; this motivates testing CoAP's efficiency at scale using Powertrace in COOJA.
- **Demand for End-to-End Analysis:**
- Prior works focus on either congestion, security, or routing individually—our approach aims to unify these under one scalable simulation model.



- Reproducibility and Open Tools:
- Using COOJA + Contiki ensures results are easily reproducible, as the tools are open-source and widely used in the community.
- Security Enhancements Emerging:
- Papers like #10 introduce secure CoAP variants (LightCert4IoT), but simulation of such enhancements at scale is missing—our topic opens ground for adding security layers over baseline CoAP.
- Congestion and Reliability Issues Remain:
- Although improved CC mechanisms exist (e.g., CoCoA+ in Paper #1), there is no unified scalable testbed for diverse topologies and traffic types. Our simulation-based study fills that need.
- Flexibility and Modularity:
- COOJA allows integrating multiple layers (MAC, RPL, 6LoWPAN), so the topic supports modular testing of CoAP in different networking stacks.



MODULES AND METHODS

Sprint	Duration	What We'll Do	Tools/Actions Used	Expected Output
1	Week 1-2	Environment setup & basic simulation	Install Contiki-NG, setup Cooja	Run basic CoAP communication between 2 motes
2	Week 3-4	Network design & traffic pattern setup	Create 10+ node topology in Cooja	Generate CoAP GET/POST traffic; analyze logs
3	Week 5-6	Performance evaluation & documentation	Use Cooja logs, energy profiler	Measure latency, PDR, energy, generate graphs

**THANK
YOU**

